Peter Bhabra

Software Engineer | AI/ML & Distributed Systems

London, United Kingdom | LinkedIn

SUMMARY

Software engineer with technical leadership experience in AI infrastructure and distributed systems. Currently leading inference platform development at Doubleword. Previously built high-availability data platforms processing 1M+ files daily at CMR Surgical.

Technologies: Rust, Python, SQL, TypeScript, React, PostgreSQL, Kubernetes, Docker, Helm, Terraform, AWS, GCP, Argo CD, GitHub Actions, vLLM, SGLang, Prometheus, Grafana, OpenTelemetry.

Specialisms: LLM inference, distributed systems, GPU scheduling, object storage, performance engineering, production operations.

EXPERIENCE

Doubleword

Member of Technical Staff - 2023 to Present

Built self-hosted inference products before moving to lead the core engineering team for Doubleword's hosted inference platform.

Hosted inference platform | Late 2025 - Present

- Led a five-person engineering team responsible for Doubleword's core hosted inference platform, and ran hiring processes to grow the team.
- Owned incident response and on-call coverage for a production inference service operating 24/7.
- Developed low-latency Rust backends for real-time and asynchronous inference, including an API gateway benchmarked at 450× faster than LiteLLM.
- Implemented durable request processing backed by PostgreSQL, covering dispatch, retries, completion, and failure handling.
- Designed and built scheduling systems that connect queued inference requests to available GPU capacity.

Self-hosted inference | 2023 - Late 2025

- Built custom inference engines with scheduling and resource management, and implemented inference research including speculative decoding.
- Delivered full-stack, on-premises deployments of open-source LLMs, production APIs, and RAG applications.
- Led technical client engagements throughout B2B sales cycles and deployment.
- Built a Rust-based inference engine using Candle for Stable Diffusion models.
- Built release and deployment automation, Helm charts, and custom Kubernetes operators, and managed self-hosted CI/CD clusters.

EXPERIENCE / CONTINUED

CMR Surgical

Graduate Software Engineer - 2021 to 2023

Worked within an agile, high-performing team delivering scalable data platforms using AWS serverless microservice architecture.

- Built and maintained data platforms processing over one million files per day
- Reduced AWS monthly costs from $60,000 to $35,000 (42% reduction) within one month
- Built HTTP Live Streaming and fine-grained permissions systems from scratch using Open Policy Agent
- Led team projects including facilitating collaboration sessions, breaking down ideas into executable stories, and quarterly planning
- Worked with telemetry and video data to deliver customer insights with frontend teams
- Utilized CI/CD techniques with TeamCity and GitlabCI for reliable releases across multiple deployment environments
- Developed AWS serverless applications using microservice architecture

EDUCATION

Durham University

Master of Science · Scientific Computing and Data Analysis - First Class Honours

Durham University

Bachelor of Science · Physics - First Class Honours

SELECTED TALKS

The Massively Parallel Agent Stack

Cloud Native & Open Source AI Conference · 2026 Watch recording

Gateways Unlock AI Builders

Cloud Native AI UG · 2025 Watch recording

TECHNICAL WRITING

Writing on AI infrastructure, systems engineering, and performance. Read the blog

Gigatoken: making tokenisation care about the hardware